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Deep learning-based downscaling of ERA5-Land temperature to 250 m resolution over the Trentino–South Tyrol Alpine region

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ABSTRACT

High-resolution near-surface temperature data are essential in mountainous regions, where complex topography induces strong spatial and temporal variability. However, coarse-resolution reanalysis products such as ERA5-Land (9 km) fail to represent fine-scale thermal patterns. This study presents a UNet based deep learning framework to downscale ERA5-Land 2-m air temperature to 250 m resolution at a 6-hourly temporal frequency over the Trentino–South Tyrol Alpine region. The model is trained and validated using the observation-based ALPINE-TST-250 gridded dataset, derived from more than 300 weather stations, and integrates high-resolution elevation data along with physically meaningful auxiliary predictors. Including 2-m dew point temperature significantly improves performance, reducing the RMSE from 2.32 °C to 2.05 °C and the MAE from 1.76 °C to 1.54 °C. The resulting 2011–2021 downscaled dataset successfully reproduces sub-kilometer spatial gradients and temporal variability absent in the original ERA5-Land fields. These results highlight the potential of deep learning approaches to enhance temperature representation in complex alpine terrain for climate and environmental applications.

1. Introduction

High-resolution weather and climate prediction is essential for regions with complex topography, where elevation strongly controls local meteorological conditions. In Trentino–South Tyrol, Alps, temperature zonation is especially heterogeneous since some areas are much steeper and at higher elevation. ERA5-Land, along with other global and regional reanalysis datasets, offers large-scale climate data at 9 km resolution, but still fail to represent small-scale heterogeneities.

Despite the increasing availability of global reanalysis data, a critical gap persists in generating meteorological products that accurately capture these localized microclimates, particularly in regions of complex alpine topography. Conventional statistical downscaling often struggles to model the extreme, non-linear thermal processes inherent to these areas. New approaches utilizing machine learning have begun to enhance the spatial resolution of temperature products from satellite-derived data to better capture these fine-scale patterns (Patil et al., 2024a).

To address this, our paper introduces a novel deep learning framework that provides an unprecedented high spatio-temporal downscal-

ing of ERA5-Land 2-meter temperature. We successfully disaggregate the data from 9 km resolution to 250 m at a 6-hour frequency (four times daily), overcoming the limitations of coarser products for local-scale applications. Beyond spatial resolution enhancement, the contribution of this work lies in the formulation of an operationally feasible and physically informed deep learning framework for high-frequency temperature downscaling in complex Alpine terrain. Unlike many previous studies that focus on daily products or emphasize architectural novelty, this study targets a 6-hourly temporal resolution at 250 m spatial scale, which is critical for resolving diurnal thermal processes in mountainous environments. Furthermore, the model design explicitly balances physical relevance and computational feasibility by integrating a limited set of carefully selected predictors — elevation and 2-m dew point temperature — under realistic memory constraints. Through an ablation-based analysis, the study demonstrates how physically meaningful auxiliary variables can substantially improve near-surface temperature representation without increasing model complexity, providing a practical blueprint for regional scale operational downscaling applications.

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1.1. Background

1.1.1. Downscaling techniques in complex terrain

Several approaches have been proposed to address the challenge of coarse-resolution climate data. Statistical methods, including regression-based downscaling, have been applied with some success in mountainous areas (e.g., Misra et al., 2024; Ntagkounakis et al., 2023). However, their capability to model the highly non-linear relationships characteristic of complex topography is often limited.

More recently, deep learning has emerged as a powerful alternative, with convolutional neural networks (CNNs) demonstrating improved accuracy in spatial downscaling tasks. This shift aligns with the broader evolution of AI-assisted remote sensing, which is transitioning from basic data acquisition to advanced cognitive services and intelligent decision-making (Xu et al., 2025). Recent advancements have explored hybrid and attention-based architectures to overcome these limits. For instance, Patil et al. (2024b,a) utilized a combination of CNNs and Extreme Learning Machines (ELM) to address scale dependency in Land Surface Temperature (LST) downscaling. Similarly, S. Li et al. (2025) introduced an Attention-Mechanism U-Net (AMUN) to enhance ERA5-Land products, demonstrating that integrating attention modules can more deeply extract relationships between auxiliary surface features and temperature fields.

Sha et al. (2020a,b) applied U-Net for downscaling meteorological variables in complex terrain, while Zhang et al. (2022) demonstrated the effectiveness of downscaling hourly air temperature simulations over complex topography. Furthermore, Jiang et al. (2021), Lin et al. (2023), Loganathan et al. (2024), and Sebbar et al. (2023), Schmidt and Ludwig (2024), Chen et al. (2025) further showed that deep learning can outperform traditional approaches in capturing local climate variability. The U-Net architecture, initially developed for image segmentation, is particularly favored due to its skip connections, which are crucial for combining low-resolution contextual information from the encoder with high-resolution spatial details in the decoder (Ryu et al., 2025). This design choice has been empirically proven to improve downscaling performance in mountainous regions (Kumar et al., 2025). Furthermore, U-Net variants are actively used in weather forecasting for interpolation and short-term prediction (Wang, 2024), and its enhanced forms, like the Residual U-Net, are explicitly leveraged for temporal downscaling of geophysical gridded data (Wang et al., 2024). More advanced approaches even adapt the probabilistic U-Net structure to statistical climate downscaling (Alipourhajiagha et al., 2025).

Building on this foundation, the present study proposes a U-Net-based model to downscale ERA5-Land air temperature from 9 km to 250 m resolution, using the ALPINE-TST-250 dataset as a reference.

1.1.2. Rationale for auxiliary predictors

To improve accuracy and spatial representation, we incorporate elevation and dew point temperature as auxiliary predictors, as suggested in Kumar et al. (2025) and successfully used in previous studies (Sha et al., 2020a; Bhakare et al., 2025). Specifically, we included 2-m dew point temperature ($d2m$) as an auxiliary predictor because it provides a direct, physically meaningful measure of near-surface moisture that is tightly linked to temperature through boundary-layer thermodynamics. It helps the model distinguish humid valley environments from drier slopes and improves representation of diurnal and nocturnal temperature contrasts, which is critical in mountainous areas. Empirical studies and global validations show robust, strong relationships between near-surface air temperature and dew point (e.g., Famiglietti et al.; Ali et al., 2018), and previous downscaling and reanalysis-evaluation studies commonly treat $d2m$ as a relevant predictor for improving near-surface temperature fields (Stensrud et al., 2003).

Regarding the potential use of satellite-derived Land Surface Temperature (LST) and vegetation indices (VI) as additional predictors, We acknowledge that air temperature exhibits strong variability across land cover types, as vegetation can substantially modulate ground

surface temperature through infrared radiation effects (J. Li et al., 2025). Furthermore, effective gap-filling methodologies exist that can partially mitigate cloud-cover data losses in satellite-derived LST products (Li et al., 2023). However, several fundamental constraints preclude their adoption in the present framework. First, our downscaling product operates at a 6-hourly temporal frequency continuously over 11 years (2011–2021), yielding approximately 15,460 time steps. In Alpine terrain, cloud contamination of MODIS LST frequently exceeds 60%–70% (Li et al., 2023), meaning that gap-filling would need to be applied to a majority of time steps, introducing compounding reconstruction uncertainties that are difficult to quantify and would compromise the physical integrity of the training signal. Second, and more critically, LST represents skin temperature, which diverges systematically from 2-m air temperature (our target variable) in Alpine terrain due to aspect, shading, slope angle, and surface thermal inertia effects; thus LST constitutes a physically inconsistent predictor for T2m downscaling. Our choice of ERA5-derived $d2m$ as an all-weather, gap-free atmospheric moisture proxy is therefore both physically and operationally more appropriate for this specific task. Nevertheless, we recognize that incorporating gap-filled LST as a post-processing refinement step represents a promising avenue for future work (see Section 4.2).

Our primary contributions are therefore threefold:

- The development of a robust U-Net architecture specifically tuned to capture sub-kilometer temperature variability and thermal biases within the highly complex and non-linear microclimates of the Alpine domain.
- The provision of the first validated temperature downscaling product over this specific region at the 250-meter resolution with a 6-hour temporal step, establishing a new benchmark for high-frequency climate analysis.
- A demonstration of how the strategic integration of topographic and physical predictors (Elevation and Dew Point Temperature) significantly enhances the model's ability to reproduce fine-scale patterns, leading to a substantial improvement in prediction accuracy.

The remainder of this paper is organized as follows: Section 1 provides the necessary background on downscaling techniques and predictor selection. Section 2 details the data and methodology, followed by results in Section 3 and a discussion in Section 4, and finally a conclusion in Section 5.

2. Materials and methods

2.1. Study area

The study focuses on Trentino–South Tyrol, a mountainous region in northeastern Italy characterized by steep elevation gradients, complex terrain, and pronounced spatial and temporal variability in temperature. Such alpine environments are particularly sensitive to climate variability and pose significant challenges for both weather forecasting and climate downscaling, making them priority regions for high-resolution studies. This explains the growing research interest in the area; for example, Crespi et al. (2021) developed a high-resolution daily temperature dataset at 250 m over Trentino–South Tyrol, via station interpolation, providing a benchmark for fine-scale climate variation. Bhakare et al. (2024) compared ML methods (ANN, RF, CNN) for downscaling ERA5-Land to 1 km over Italian Alpine valleys, showing CNN performs best in many seasons.

2.2. Datasets

We used three primary datasets:

- **ERA5-Land:** hourly 2-m temperature (t_{2m}) and dew point temperature (d_{2m}) at 9 km resolution for the period 2011–2021. ERA5-Land is widely employed as a benchmark dataset in several downscaling studies (Niazkar et al., 2024; S. Li et al., 2025).
- **ALPINE-TST-250:** a high-resolution (250 m) dataset of hourly 2-m temperature over Trentino–South Tyrol, generated from more than 300 ground stations using kriging with elevation as a covariate (Dhawan et al., 2025) (downloaded 30/05/2025).
- **Elevation:** Shuttle Radar Topography Mission (SRTM) 30 m digital elevation model, aggregated to 250 m via block averaging to match the ALPINE grid.

We additionally included **dew point temperature** (d_{2m}) as an auxiliary predictor alongside t_{2m} , given its strong physical correlation with near-surface moisture and its relevance for capturing nocturnal temperature minima and valley microclimates (Famiglietti et al.; Ali et al., 2018).

2.3. Data preprocessing and alignment

The study utilizes data spanning an eleven-year period, from 2011-01-01 to 2021-12-31, for both the coarse ERA5-Land inputs and the ALPINE-TST-250 target dataset. This extensive temporal coverage ensures the model is trained on a wide range of meteorological conditions characteristic of the Alpine environment:

1. **Elevation grid:** CSV elevation data were reshaped into a 2D latitude–longitude grid and upsampled from 30 m to 250 m.
2. **Coordinate Transformation:** ALPINE-TST-250 temperature data, originally in Easting/Northing coordinates, were converted to standard latitude/longitude coordinates (EPSG:3035 → EPSG:4326) using `pyproj` to ensure consistency with the ERA5-Land grid.
3. **Temporal Consistency:** To reduce storage and focus on key meteorological periods, the ALPINE hourly data (2011–2021) were subsampled, retaining only four daily time steps (00:00, 06:00, 12:00, 18:00 UTC), corresponding to the 6-hourly temporal frequency of the final downscaling product.
4. **Spatial Regridding:** ERA5-Land variables were initially extracted; however, due to memory constraints, only t_{2m} and d_{2m} were retained and regridded onto the 250 m resolution ALPINE target grid via bilinear interpolation using the following code snippet: `era5.interp(latitude=alpine.latitude, longitude=alpine.longitude, method="linear")`. This ensured perfect spatial alignment between the coarse input and the fine target.
5. **Missing Data Handling:** Small missing gaps in time or space within the ERA5 data were filled using linear interpolation (`interpolate_na`) along the time dimension. Remaining NaNs, typically located outside the ALPINE coverage area, were handled by applying a spatial mask derived from the initial ALPINE dataset (`mask = ~np.isnan(alpine.isel(time=0))`), ensuring all valid regions were consistent across both input and target data.
6. **Feature Selection and Format:** Due to memory constraints (≈ 41 – 51 GB per year), only t_{2m} , d_{2m} , and *elevation* were retained as predictors. The final datasets were normalized using Z-score standardization and saved in Zarr format (`X.to_zarr("ERA5_prepared.zarr")`) aligned in space and time, with dimensions (time, latitude, longitude, variable), which allows for efficient chunked streaming and avoids full loading into RAM during training.
7. **Data Splits:** The final data were partitioned for machine learning: Training (2011–2018), Validation (2019), and Testing (2020–2021).

Our feature set X is composed of three channels, defining the input tensor shape before cropping as (3, 15460, 651, 648) (variables, time, latitude, longitude), where the 15460 time steps correspond to 10 years of 6-hourly data (plus one leap year). The input variables are: (1) 2-meter temperature (T_{2m}), (2) Elevation (static auxiliary predictor), and (3) 2-meter dew point temperature (d_{2m}), which was introduced in the second model configuration for ablation analysis.

As shown in Fig. 1, the coarse ERA5-Land fields fail to capture the fine-scale temperature variability evident in the ALPINE dataset.

2.4. Model architecture and input design

U-Net Architecture

We adopt the U-Net architecture as the core model for the downscaling framework owing to its strong performance in image-to-image regression tasks and its ability to preserve fine-scale spatial information. Originally proposed for biomedical image segmentation (Ronneberger et al., 2015), U-Net has since been widely applied to geoscientific problems where multi-scale spatial dependencies are essential, including precipitation estimation, climate super-resolution, and temperature downscaling in complex terrain (Sha et al., 2020a; Stengel et al., 2020; Kumar et al., 2025). These studies consistently demonstrate that U-Net outperforms traditional super-resolution CNNs and shallow architectures in capturing both coarse-scale atmospheric structure and localized terrain-driven variability.

A key strength of U-Net lies in its encoder–decoder structure with skip connections, which enables the model to merge deep semantic features with high-resolution spatial details. In the context of Alpine meteorological downscaling, this architectural property is crucial: the encoder learns large-scale synoptic patterns from reanalysis data, while the skip connections reconnect these representations with fine-grained features relevant to temperature variability induced by topographic complexity. Prior research has shown that such multi-scale feature fusion improves the reconstruction of sharp gradients and local extremes, especially in heterogeneous mountainous regions (Vandal et al., 2018; Kumar et al., 2025).

In this study, we implement a U-Net following the general design of Ronneberger et al. (2015) (Fig. 2). The model takes as input a tensor of shape (t, C, H, W) , where C is the number of channels. Both dynamic and static predictors are included:

- ERA5 2-m temperature (t_{2m}) (dynamic)
- ERA5 2-m dew point temperature (d_{2m}) (dynamic)
- Elevation (static, resampled to 250 m resolution)

The combination of dynamic atmospheric predictors with high-resolution static topography allows the model to leverage both meteorological and physiographic controls on near-surface temperature. The output is a tensor of shape $(t, 1, H, W)$ representing the downscaled 2-m air temperature field at 250 m resolution.

Overall, the choice of U-Net is motivated by its demonstrated ability to (i) capture multi-scale atmospheric–terrain interactions, (ii) reconstruct sharp spatial gradients that are critical in Alpine environments, and (iii) outperform conventional super-resolution approaches in geophysical downscaling tasks.

The parameters presented in Table 1 were selected to ensure stable convergence while maintaining computational feasibility for large-scale, long-term regional downscaling.

2.5. Training setup and evaluation

- **Preprocessing:** Input grids were cropped to (648×648) to ensure divisibility by 8 for pooling/upsampling operations within the U-Net. Data were normalized, and spatial masks were applied for missing values.

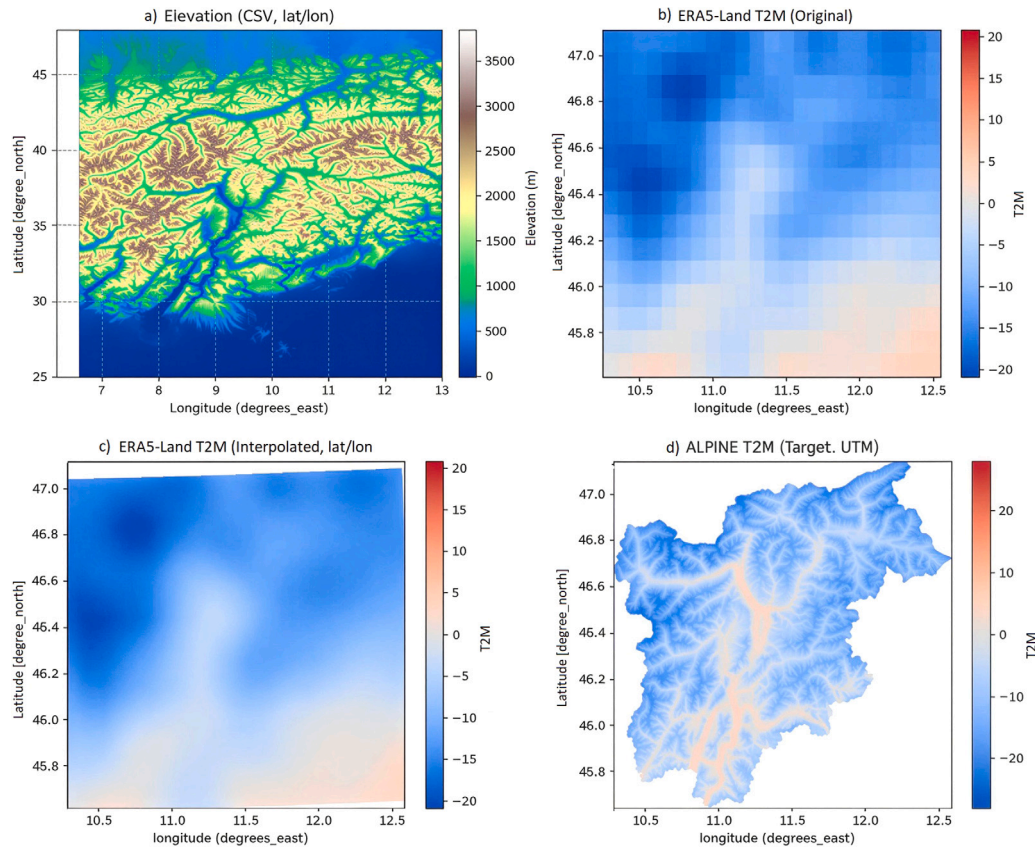


Fig. 1. Comparison of datasets used in the downscaling framework over the Trentino-South Tyrol region. (a) Elevation at 250 m resolution (SRTM, upscaled from 30 m). (b) ERA5-Land original 2-m air temperature (9 km). (c) ERA5-Land 2-m temperature interpolated onto the ALPINE grid. (d) ALPINE-TST-250 target 2-m temperature at 250 m resolution. The figure illustrates the resolution gap between coarse ERA5 reanalysis and fine-scale ALPINE observations.

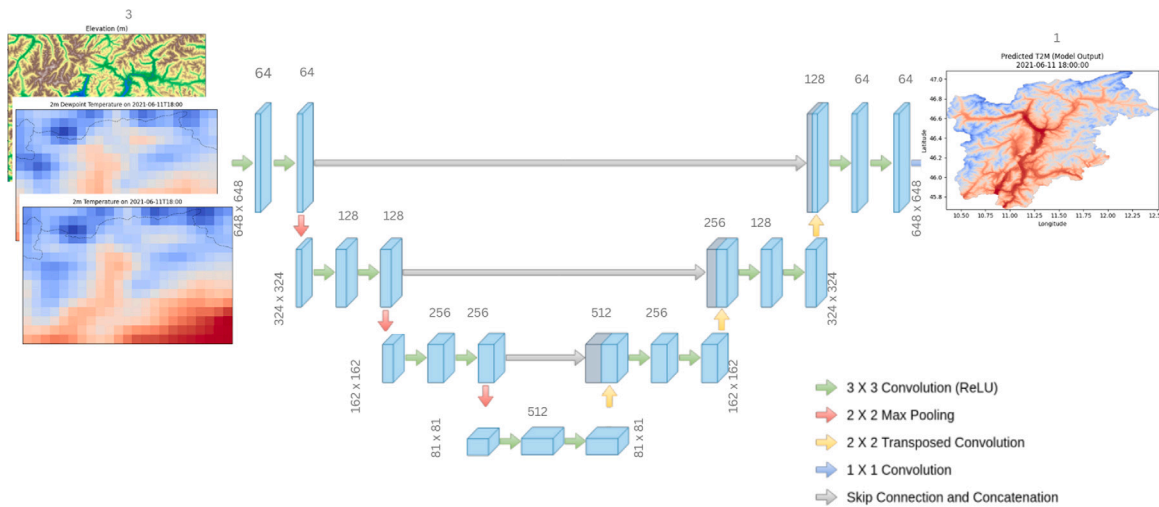


Fig. 2. U-Net architecture for topographic downscaling of near-surface temperature. The model integrates dynamic atmospheric predictors (ERA5 2m temperature and dew point) with high-resolution static elevation data (250 m) to predict fine-scale temperature fields. The encoder-decoder structure with skip connections (gray arrows) facilitates the reconstruction of sharp spatial gradients across Alpine terrain. Numerical values indicate feature map channels (e.g., 64, 128) and spatial dimensions (e.g., 648 × 648).

Table 1

U-Net model architecture and training hyperparameters.

Parameter	Value
Architecture	U-Net
Input channels	2 (baseline), 3 (enhanced)
Output channels	1
Loss function	Masked Mean Squared Error
Optimizer	Adam
Batch size	16
Number of epochs	40 (early stopping)
Input patch size	648 × 648
Normalization	Z-score standardization

Table 2

Computational environment and software used in this study.

Component	Specification
GPUs	2 × NVIDIA RTX 3090 (24 GB VRAM each)
CPU	Linux-based workstation
System memory	251 GB RAM
Deep learning framework	PyTorch
Data format	Zarr
Operating system	Linux

- **Loss Function:** The model was trained using the masked Mean Squared Error (MSE) loss function.
- **Optimizer:** The Adam optimizer was employed with a batch size of 16.
- **Metrics:** Performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).
- **Hardware and Duration:** Training was performed on dual NVIDIA RTX 3090 GPUs (24 GB VRAM each) within a Linux environment (251 GB RAM). One epoch required approximately 7 min, with the training process stopping at epoch 40 (using early stopping), resulting in a total training duration of ≈4 h 40 min.
- **Diagnostics:** Final assessment included spatial maps, difference maps, and time series comparisons to evaluate model performance across different regions and temporal periods.

Training and computational setup

The model was trained using PyTorch, leveraging the Zarr format for efficient large-scale data handling. The full dataset was processed into 15,460 cropped training samples, resulting in final input and target training array shapes of X_{cropped} : (15460, 3, 648, 648) and Y_{cropped} : (15460, 648, 648), respectively. The model training was significantly accelerated by utilizing two NVIDIA GeForce RTX 3090 GPUs with DataParallel, confirming the feasibility of deploying this approach for efficient regional downscaling.

The experimental environment is summarized in Table 2.

3. Results

3.1. Baseline model ($t2m + \text{elevation}$)

The baseline U-Net model, trained using ERA5 2-m air temperature and elevation, achieved an RMSE of 2.32 °C and an MAE of 1.76 °C. Overall, the model successfully reproduced the dominant spatial temperature patterns across the study domain, although it tended to underestimate temperature extremes in areas characterized by strong elevation gradients.

Figs. 3–8 collectively summarize the spatial, temporal, and statistical behavior of the baseline configuration. The downscaled fields exhibit a clear improvement over both the native and interpolated ERA5-Land products, effectively narrowing the resolution gap with respect to the ALPINE reference dataset. Difference maps highlight that

the largest residual errors are primarily confined to high-altitude and steep-terrain regions, where thermal variability is most pronounced.

Model training and validation diagnostics indicate stable convergence and satisfactory generalization, with no evidence of overfitting. Time-series comparisons further demonstrate that the model preserves temporal consistency at representative locations, despite a slight underestimation of extreme temperature values.

The spatiotemporal error maps (Fig. 6) reveal a systematic diurnal pattern in residual errors, presented in strict temporal order from 00:00 to 18:00 UTC. At 00:00 UTC, warm biases are concentrated in high-elevation plateaus, likely associated with the model's difficulty in capturing nocturnal radiative cooling over snow-covered or rocky surfaces where the temperature–elevation lapse rate departs from its daytime mean. At 06:00 UTC, cold biases emerge prominently in deep valley floors, a signature consistent with cold air pooling (CAP) processes that are moisture-dependent and therefore only partially captured by the baseline model which lacks a moisture predictor. By 12:00 UTC, errors are more spatially diffuse and substantially reduced in magnitude, reflecting the model's stronger performance during convectively mixed daytime conditions when the temperature lapse rate is more predictable from elevation alone. The 18:00 UTC panel shows a transition pattern with moderate residuals at mid-elevations, linked to the onset of evening boundary-layer decoupling.

3.2. Extended model ($t2m + d2m + \text{elevation}$)

Incorporating 2-m dew point temperature as an additional predictor resulted in a clear improvement in model performance, reducing the RMSE to 2.05 °C and the MAE to 1.54 °C. The enhanced configuration more accurately captured fine-scale spatial gradients and temperature extremes, particularly in complex mountainous areas.

Figs. 9–14 present a comprehensive comparison of the extended model results. Relative to the baseline case, spatial error patterns exhibit reduced magnitudes, and prediction–target differences are consistently lower across all analyzed times of day. The improved performance reflects the additional physical information provided by dew point temperature, enhancing the representation of moisture-related thermal contrasts.

Training and validation diagnostics further confirm the robustness of the enhanced model. Both loss curves exhibit a smooth and consistent decrease without pronounced oscillations, indicating stable convergence throughout training. In contrast to the baseline configuration (Fig. 7), which reached its best validation performance early (best validation loss = 0.0665 at epoch 10; final validation loss = 0.0690) and subsequently showed increased variability, the extended model continued to improve steadily, achieving its optimal validation loss of 0.0508 at epoch 39, with a final validation loss: 0.0512 (Fig. 13). The final test metrics (RMSE = 2.0456 °C, MAE = 1.5384 °C) closely match the validation performance, confirming strong generalization and the absence of overfitting. By comparison, the baseline model exhibits higher test errors (RMSE = 2.3178 °C, MAE = 1.7646 °C) and greater validation loss fluctuations after the early training phase, highlighting the stabilizing contribution of dew point temperature.

The spatiotemporal error maps of the extended model (Fig. 12) confirm the improvement brought by $d2m$ across all four UTC time steps (00:00, 06:00, 12:00, 18:00), presented in strict temporal order. The most pronounced improvement is observed at 06:00 UTC, where the cold bias in valley floors — a signature of cold air pooling — is substantially reduced relative to the baseline (Fig. 6). This improvement is physically expected: $d2m$ provides direct information on near-surface moisture content, which governs nocturnal longwave emission and the depth of cold air pools. At 00:00 UTC, the warm bias over high-elevation plateaus is also attenuated, as $d2m$ implicitly encodes cloud fraction information that modulates the nocturnal surface radiative budget. At 12:00 and 18:00 UTC, the difference between baseline and extended error maps is smaller, consistent with the reduced role of

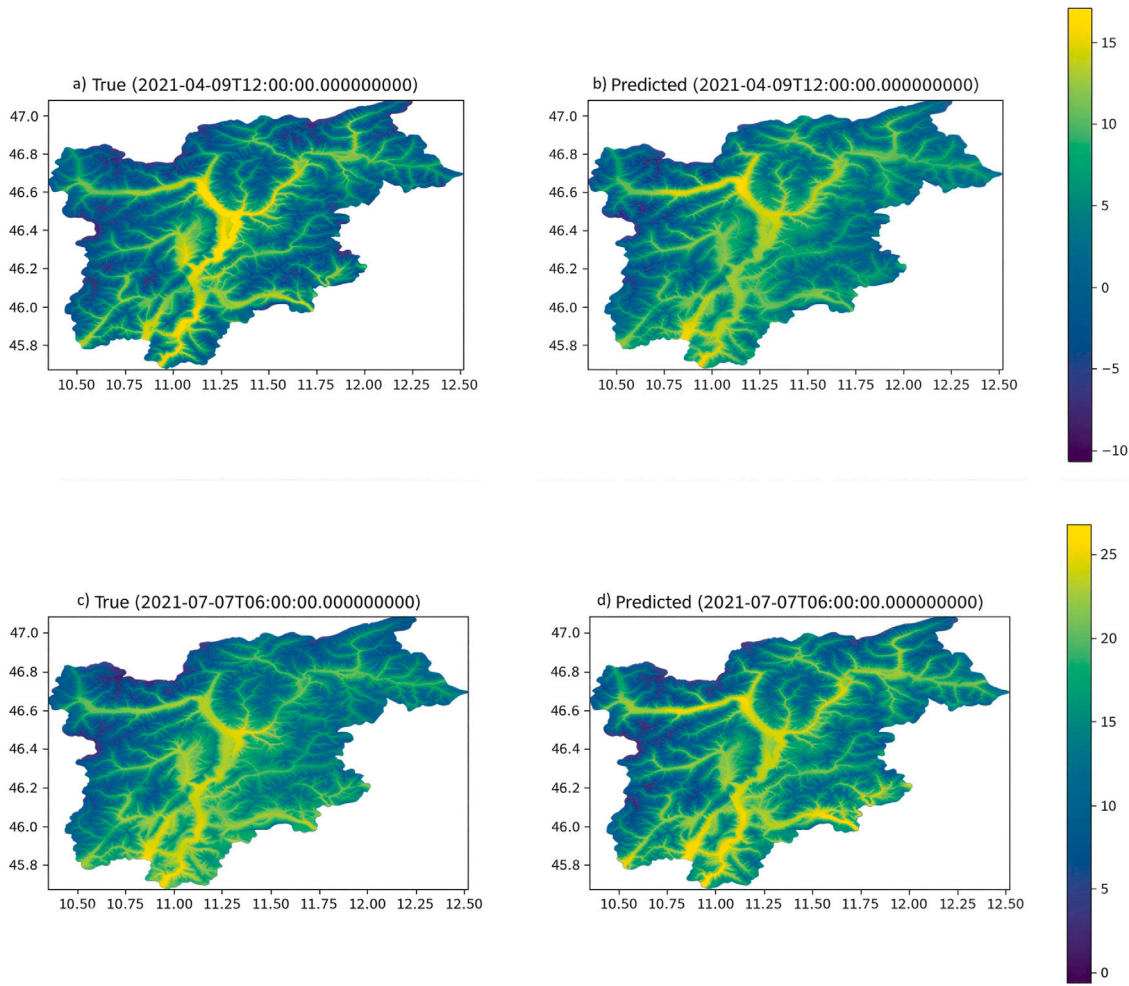


Fig. 3. Comparison of observed (ALPINE) vs. predicted (U-Net) 2-m temperature for the Baseline model. Panels are arranged horizontally for each selected time step: (a) and (c) show ALPINE observations; (b) and (d) show corresponding U-Net predictions using only t_{2m} and elevation as inputs. A shared temperature colorbar applies within each row (identical for the True and Predicted panels of that row), enabling direct True-vs-Predicted comparison. Note that the two rows use different colorbar ranges reflecting their different seasonal time steps; colorbar limits in this figure match those in the corresponding rows of (Fig. 9) for cross-model comparison. While the model captures the primary topographic controls, the spatial heterogeneity reflects a simplified relationship between elevation and temperature over the Trentino–South Tyrol region at 250 m resolution. Note the systematic underestimation in high-elevation valleys, which is addressed by the extended model (Fig. 9).

moisture in regulating daytime convective lapse rates. The residual errors that persist across all time steps in both models are concentrated in the highest-elevation zones (above 2000 m a.s.l.), which are subject to greater uncertainty in the ALPINE-TST-250 reference dataset due to sparse station coverage (see Section 4.3).

3.3. Performance metrics and ablation study

To quantify the impact of auxiliary physical information, we conducted an ablation study comparing two model configurations. Model 1 (Baseline) used T_{2m} and Elevation as input, and Model 2 (Enhanced) added d_{2m} as the third input channel.

The results show a marked improvement with the inclusion of the dew point temperature, confirming its importance in capturing micro-scale thermodynamic processes:

- **Baseline Model (Model 1):** Achieved an RMSE of 2.32 °C and an MAE of 1.76 °C. Training reached stable loss values after ≈ 17 epochs (Final Validation Loss: 0.0690).
- **Enhanced Model (Model 2):** Achieved a significantly reduced RMSE of 2.05 °C and an MAE of 1.54 °C. This configuration

Table 3

Performance comparison of U-Net model configurations on the test period (2020–2021).

Model	Input variables	RMSE (°C)	MAE (°C)	Relative RMSE reduction
Model 1	t_{2m} + elevation	2.32	1.76	–
Model 2	t_{2m} + d_{2m} + elevation	2.05	1.54	11.6%

demonstrated rapid convergence, reaching superior stability after ≈ 40 epochs (Final Validation Loss: 0.0512).

The inclusion of d_{2m} resulted in an 11.6% reduction in RMSE and a 12.5% reduction in MAE. This gain demonstrates that d_{2m} is not merely an incremental feature but provides critical, non-linear information essential for resolving localized humidity and cloud cover effects that profoundly influence Alpine surface temperature.

Table 3 quantitatively summarizes the impact of incorporating dew point temperature, highlighting a consistent reduction in both RMSE and MAE relative to the baseline configuration.

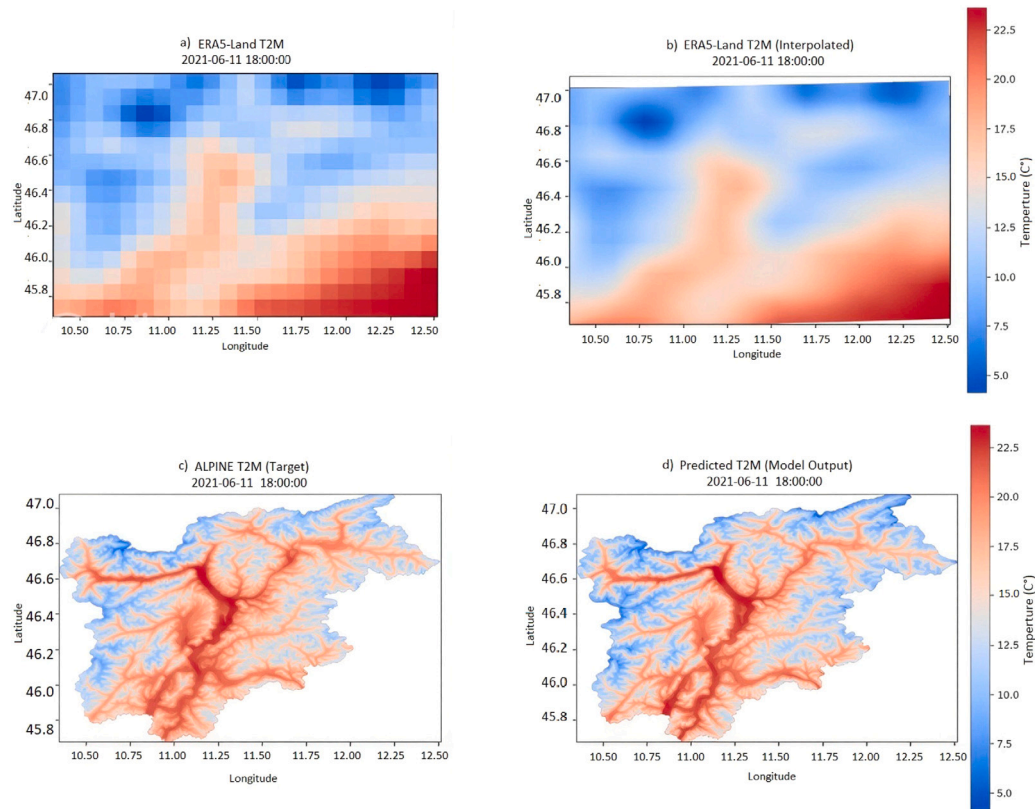


Fig. 4. Multi-scale comparison of 2-m temperature datasets over the Trentino–South Tyrol region. Panel (a) shows the original coarse-resolution ERA5-Land (9 km), while (b) illustrates the ERA5-Land field interpolated to the ALPINE grid. Panel (d) accurately recovers the high-frequency topographic details present in the 250 m ALPINE-TST-250 reference (c), demonstrating the baseline model's ability to super-resolve complex Alpine features. All panels use an identical colorbar range to enable direct comparison with the extended model (Fig. 10).

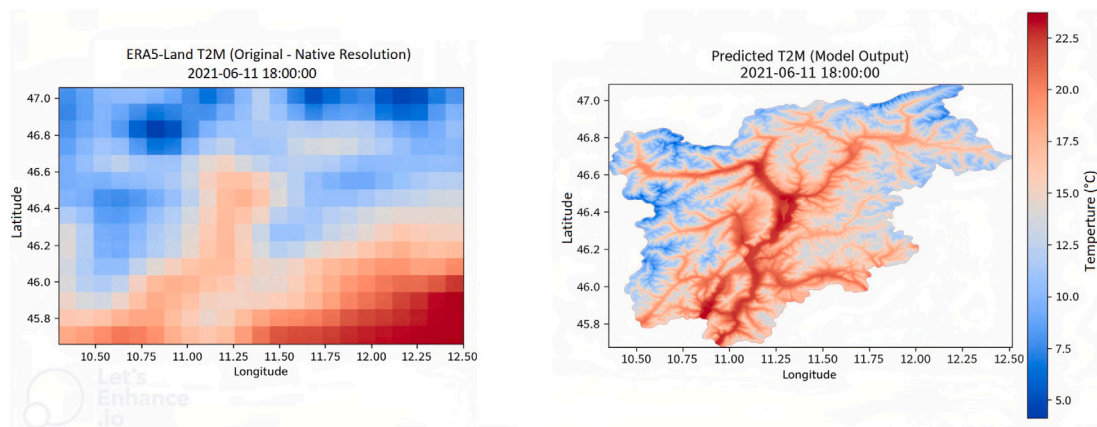


Fig. 5. Spatial comparison between coarse-resolution ERA5-Land (9 km; left) and high-resolution U-Net predictions (250 m; right) of 2-m air temperature over Trentino–South Tyrol for a representative summer afternoon time step (2021-06-11 18:00 UTC). The stark contrast in spatial detail illustrates the downscaling gain achieved by the U-Net framework. Readers should focus on the fine-scale valley–ridge temperature gradients resolved in the right panel, which are absent in the coarse ERA5-Land field on the left.

Visual analysis

The visual analysis of the predicted maps (Figs. 3, 9) confirms that U-Net predictions closely align with ALPINE observations. Time series comparisons (Figs. 8, 14) indicate small biases, with a slight tendency toward overestimation during cold periods.

A direct comparison of the error maps (Figs. 6, 12) shows that the inclusion of *d2m* effectively mitigates the systematic cold biases observed in high-elevation valleys during the 06:00 UTC transition.

The visual comparison of multi-scale temperature fields (Figs. 4–10) demonstrates the ability of both models to recover fine-scale Alpine

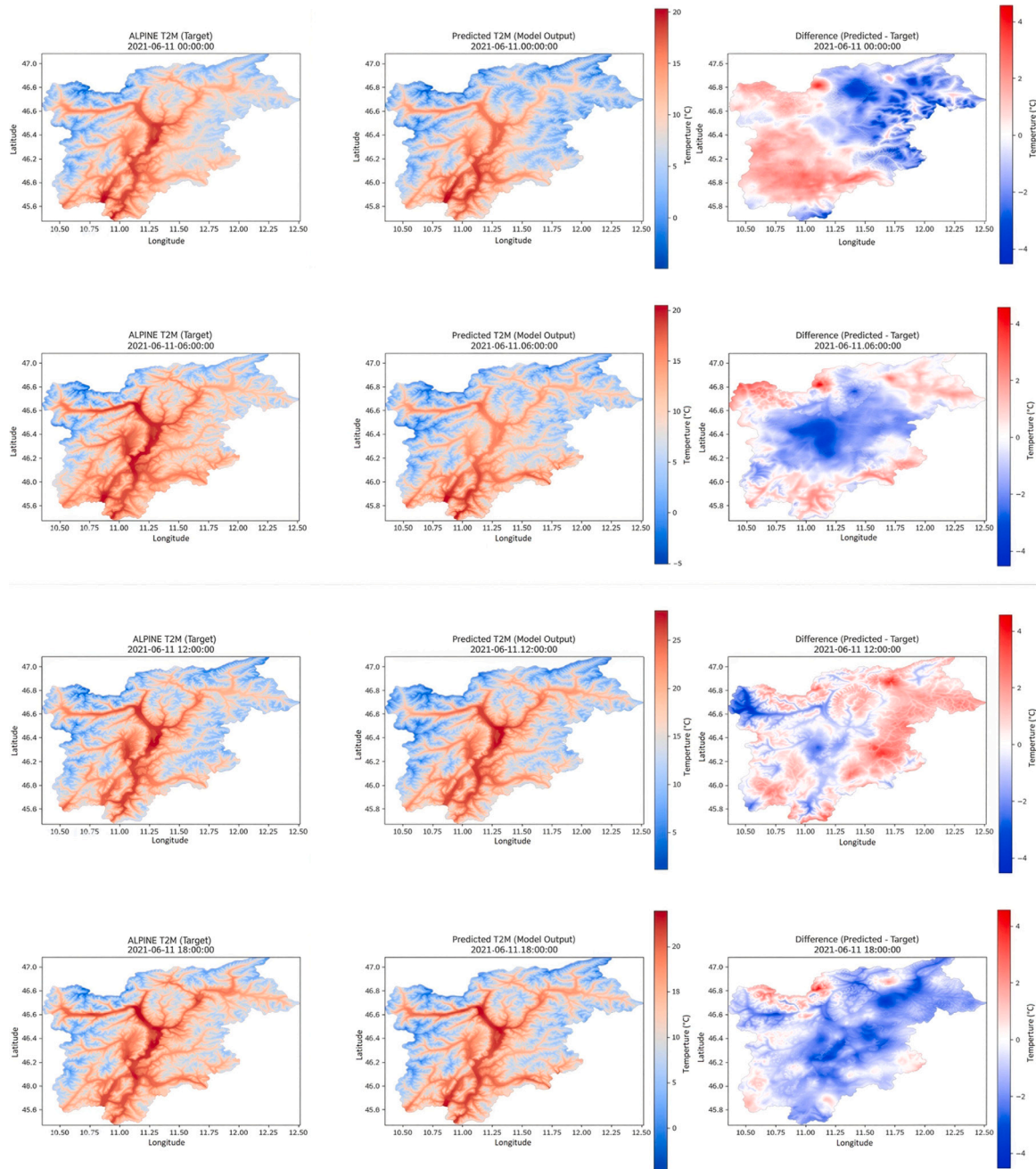


Fig. 6. Baseline U-Net spatiotemporal performance. Target (left), predictions (center), and residual maps (right, Predicted – Target). Rows 1–4: 00:00, 06:00, 12:00, 18:00 UTC (strict order). Shared residual colorbar: ± 4 °C. Warm biases at 00:00 UTC reflect nocturnal cooling underestimation over plateaus; cold biases at 06:00 UTC indicate cold air pooling in valley floors; errors are smallest at 12:00 UTC. See text and Fig. 12.

temperature structures from coarse ERA5-Land inputs. High-resolution outputs (Figs. 5, 11) further show that the extended configuration produces sharper topographic gradients and improved spatial consistency. Nonetheless, overall spatial and temporal patterns were accurately reproduced.

4. Discussion

4.1. Interpretation of high-resolution skill

Unlike previous datasets that provide daily resolution or coarser spatial products, this study delivers a validated 250 m temperature

dataset at a 6-hourly temporal resolution, enabling the analysis of diurnal thermal processes in complex Alpine terrain. The U-Net architecture demonstrated strong skill in reconstructing fine-scale temperature patterns, validating its efficacy as a super-resolution method for down-scaling in complex terrain. This success is primarily attributed to the network's skip connections, which effectively merge the coarse-scale atmospheric context provided by the ERA5 input with the high-frequency spatial details driven by the elevation data.

The contribution of dew point temperature ($d2m$) is crucial and physically meaningful. The quantitative improvement (RMSE decreased from 2.32 °C to 2.05 °C and MAE from 1.76 °C to 1.54 °C) reflects the predictor's ability to represent moisture-controlled thermal contrasts.

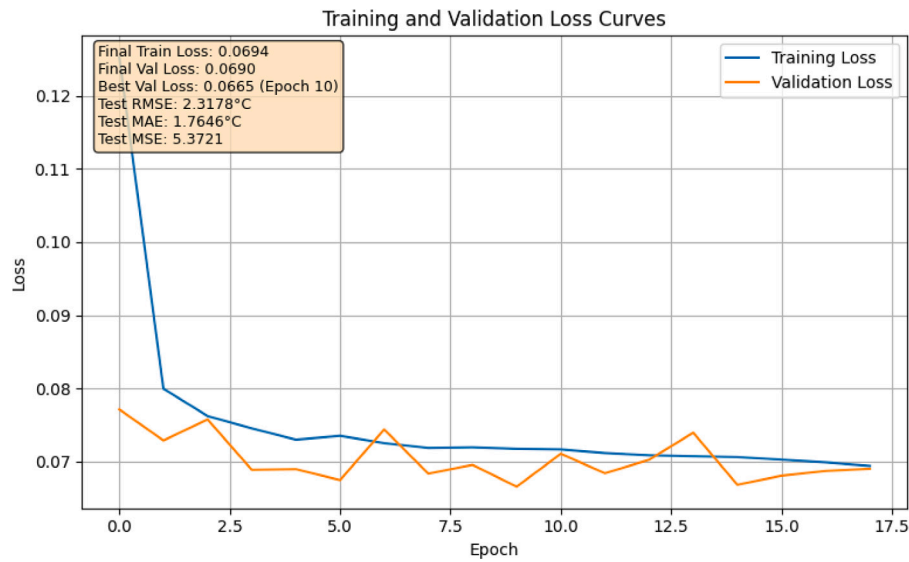


Fig. 7. Training and validation loss curves of the baseline U-Net downscaling model (Model 1: $t2m$ + elevation). Final test metrics (RMSE = 2.32 °C, MAE = 1.76 °C) are reported in the legend, demonstrating model generalization to unseen data (test period: 2020–2021).

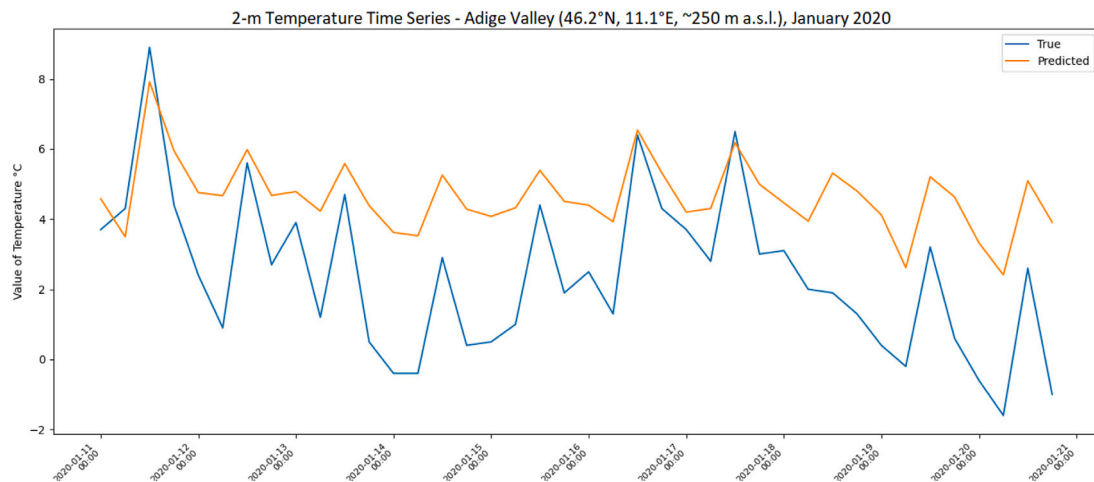


Fig. 8. Time series of observed (ALPINE, blue) and predicted (U-Net, orange) 2-m air temperature at a representative location over a 10-day period (January 2020), illustrating temporal consistency and predictive accuracy. The representative location corresponds to pixel ($i=0$, $j=182$) in the ALPINE-TST-250 grid, situated at approximately 46.2°N, 11.1°E at an elevation of ≈ 250 m a.s.l., representative of a valley-floor thermal regime in the Adige Valley (Trentino). This low-elevation valley site was selected because it exhibits the full diurnal temperature range and is representative of the populated areas where high-resolution temperature data are most needed for climate impact assessments.

As evidenced by the spatial maps, $d2m$ provides a proxy for near-surface moisture, which modifies radiative fluxes and boundary-layer processes. This enhanced information allows the model to better resolve second-order, micro-scale phenomena like Cold Air Pools in the humid valley environments, improving the distinction between cool valley floors and drier slopes, thus explaining the reduced discrepancies observed in mountainous regions. These findings are consistent with the increasing body of literature that advocates for the use of CNN-based methods and physically relevant predictors in downscaling complex terrain (Sha et al., 2020a; Lin et al., 2023; Sebbar et al., 2023). Residual errors may partly reflect uncertainties inherent in station interpolation within the ALPINE-TST-250 dataset, particularly in data-sparse high-elevation areas.

4.2. Limitations and future directions

The main limitation of this study lies in the restricted set of dynamic atmospheric predictors used for downscaling. Although elevation and

$d2m$ proved highly effective, the exclusion of variables such as wind components ($u10$, $v10$) and surface pressure (sp) due to computational constraints limited the network's access to important information on atmospheric circulation and pressure-driven variability. Integrating these dynamic predictors may further reduce errors by modeling non-static effects like advection in complex mountainous terrain.

Furthermore, while the 6-hourly temporal resolution significantly advances the spatial detail, it is a trade-off that does not fully resolve the continuous diurnal cycle. Future work should therefore explore hybrid architectures that combine the spatial power of CNNs with recurrent models (e.g., ConvLSTM) to better represent temporal dependencies and achieve truly continuous hourly resolution.

Regarding the incorporation of satellite-derived Land Surface Temperature (LST) and vegetation indices, we acknowledge that the current predictor set is intentionally minimal due to the operational constraints described in Section 1.1.2. Nevertheless, future work could explore the integration of gap-filled LST products as supplementary predictors, building on the advanced reconstruction methods reviewed by Li

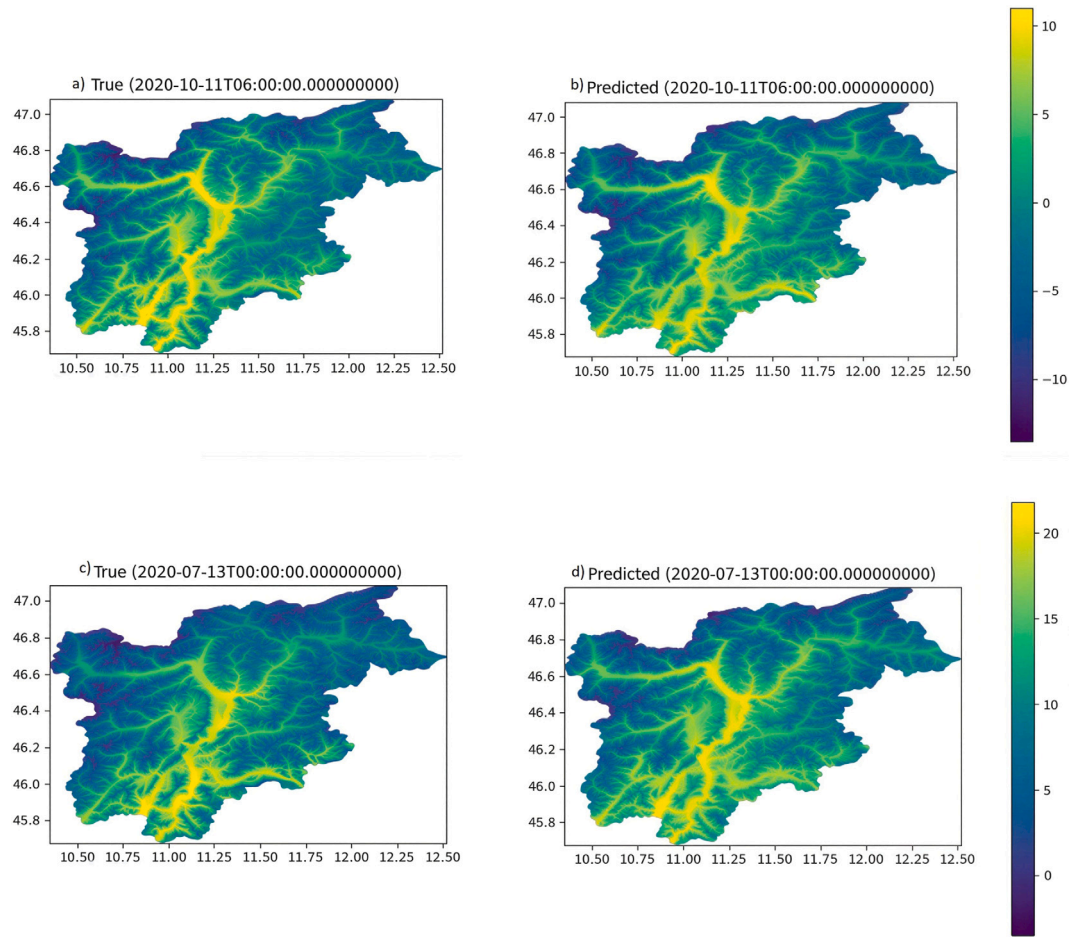


Fig. 9. Comparison of observed (ALPINE) vs. predicted (U-Net) 2-m temperature for the Full model. Panels are arranged horizontally for each selected time step: (a) and (c) show ALPINE observations; (b) and (d) show U-Net predictions using the full predictor set ($r2m$, $d2m$, and elevation). A shared temperature colorbar applies within each row (identical for the True and Predicted panels of that row), enabling direct True-vs-Predicted comparison. Note that the two rows use different colorbar ranges reflecting their different seasonal time steps; colorbar limits in this figure match those in the corresponding rows of Fig. 3 for cross-model comparison. The inclusion of dew point ($d2m$) significantly improves the representation of moisture-dependent thermal gradients and cold-air pooling in deep Alpine valleys, as visible in the sharper valley-floor temperature contrasts in panels (b) and (d).

et al. (2023), while accounting for land-cover-dependent LST variability documented by J. Li et al. (2025). Such an approach would require careful validation of gap-filling uncertainty propagation, particularly for the high cloud-cover frequencies characteristic of Alpine terrain, and the physical consistency between skin temperature (LST) and 2-m air temperature would need to be explicitly accounted for in the model design.

Finally, while promising results were obtained for the Trentino–South Tyrol region, the model’s transferability to other alpine domains and its robustness under non-stationary climate conditions requires further rigorous validation. Exploring lightweight architectures and techniques like residual learning will also be crucial to better capture temperature extremes and minimize computational demands for broader application.

The balance between computational efficiency and model complexity is detailed in Table 4, highlighting the advantages and inherent limitations of our proposed framework.

4.3. Uncertainty in the reference dataset

The ALPINE-TST-250 dataset used as reference in this study is a gridded temperature product derived from more than 300 ground stations using kriging with elevation as an auxiliary variable. While

Table 4

Trade-offs between model complexity and operational feasibility.

Aspect	Benefit	Limitation
Limited predictor set	Reduced memory usage, scalable	Missing some dynamic processes
Inclusion of d_{2m}	Improved physical realism	Dependent on ERA5 quality
6-hourly resolution	Captures diurnal variability	Not continuous hourly output
U-Net architecture	Stable, efficient training	Not architecturally novel

this dataset represents one of the most comprehensive high-resolution temperature products available for the Trentino–South Tyrol region and has been widely used in previous studies, it is not free from uncertainty. In particular, station density decreases with elevation, which may introduce larger interpolation uncertainty in high-altitude and complex terrain areas.

Consequently, part of the residual errors observed in mountainous regions may reflect uncertainty in the reference dataset rather than deficiencies in the downscaling model itself. This is an important consideration when interpreting spatial error patterns and apparent biases.

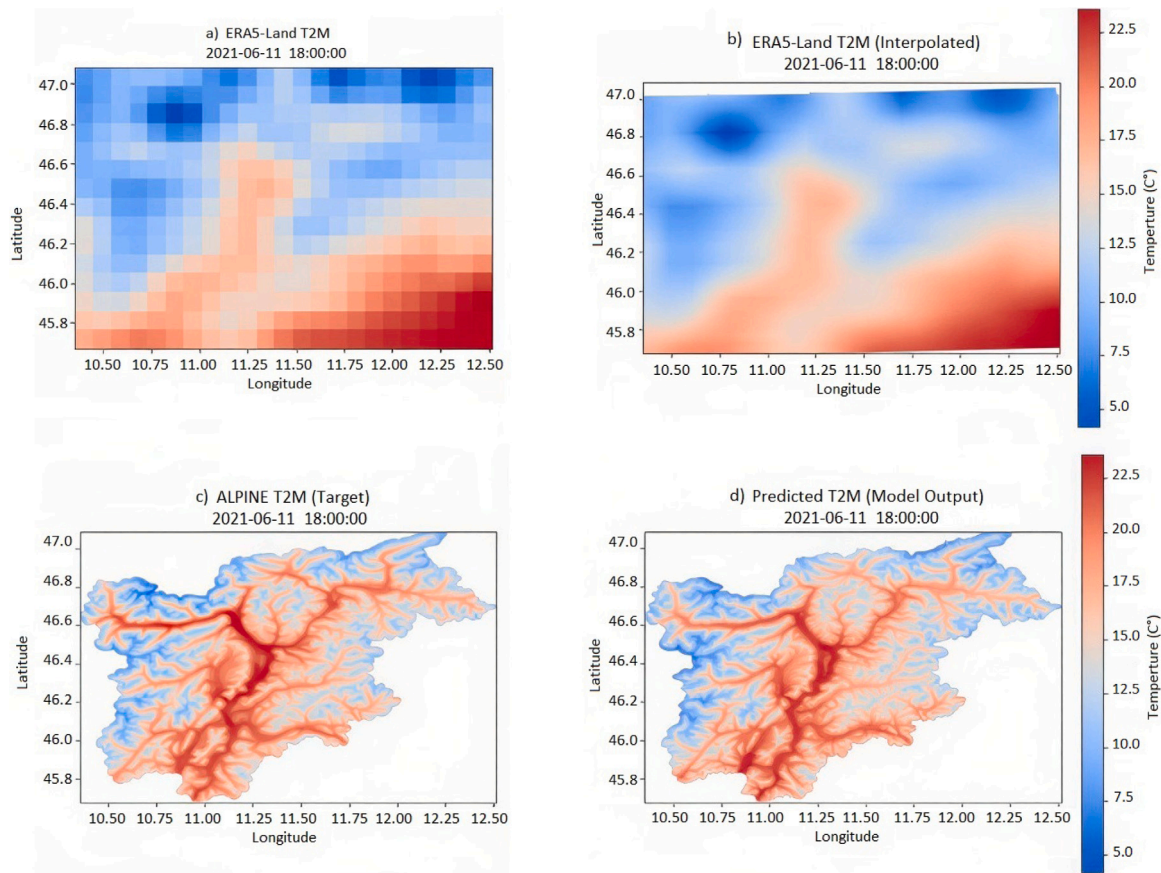


Fig. 10. Multi-scale comparison of 2-m temperature datasets over the Trentino–South Tyrol region for the extended model. Panel (a) shows the original coarse-resolution ERA5-Land (9 km), while (b) illustrates the ERA5-Land field interpolated to the ALPINE grid. Panel (d) demonstrates the extended model output, which more accurately recovers the high-frequency topographic details present in the 250 m ALPINE-TST-250 reference (c) compared to the baseline. All panels use an identical colorbar range to enable direct comparison with the baseline model.

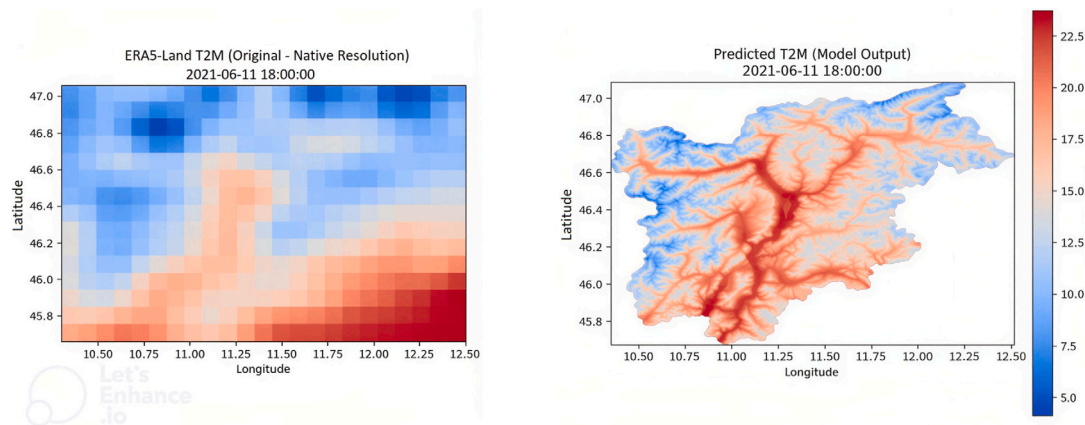


Fig. 11. Spatial comparison between coarse-resolution ERA5-Land (9 km; left) and high-resolution extended U-Net predictions (250 m; right) of 2-m air temperature over Trentino–South Tyrol for the same representative summer afternoon time step as Fig. 5 (2021-06-11 18:00 UTC). Compared to the baseline (Fig. 5), the extended model produces sharper valley–ridge gradients and improved representation of moisture-driven thermal contrasts in deep Alpine valleys.

Despite this limitation, ALPINE-TST-250 remains a suitable benchmark for evaluating spatial and temporal consistency of downscaled temperature fields at sub-kilometer resolution. Future work will focus on complementary validation using independent station observations when consistent and homogeneous datasets are available.

5. Conclusion

This research showed how a deep learning-based framework can address the pressing challenge of acquiring high spatio-temporal resolution temperature data over the complex terrain of the Trentino–South

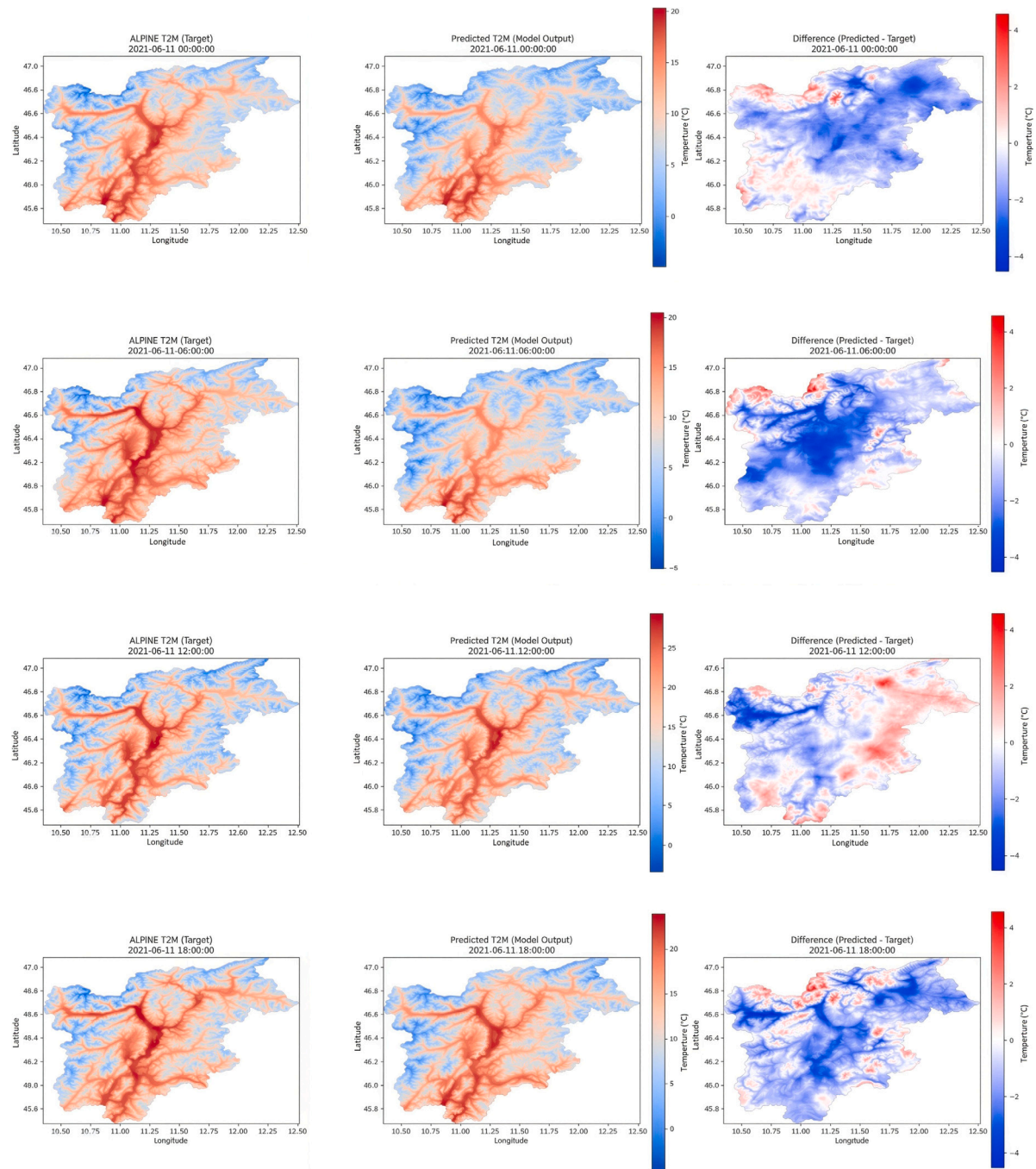


Fig. 12. Extended U-Net spatiotemporal performance. Target (left), predictions (center), and residual maps (right, Predicted – Target). Rows 1–4 correspond to 00:00, 06:00, 12:00, and 18:00 UTC respectively (strict temporal order). The shared residual colorbar (± 4 °C) is identical to Fig. 6 for direct comparison. Including $d2m$ reduces bias at all time steps; the largest improvement is at 06:00 UTC (cold air pooling). See text for full discussion.

Tyrol Alpine region. We successfully downscaled ERA5-Land 2-meter temperature from 9 km to 250 m at 6-hourly intervals. The U-Net architecture, paired with static elevation data, proved highly effective at resolving the strong temperature gradients characteristic of mountainous areas. The incorporation of dew point temperature ($d2m$) resulted in a significant enhancement in overall performance (RMSE improvement from 2.32 °C to 2.05 °C and the MAE from 1.76 °C to

1.54 °C), demonstrating the model's ability to capture subtle, micro-scale thermodynamic processes often missed by conventional methods. The resulting 250 m, 6-hourly temperature dataset for the Trentino–South Tyrol region represents a new, high-fidelity benchmark product highly valuable for local climate impact assessments, hydrological modeling, and risk management in this climatically sensitive region. Future research should aim at extending this framework to a broader range

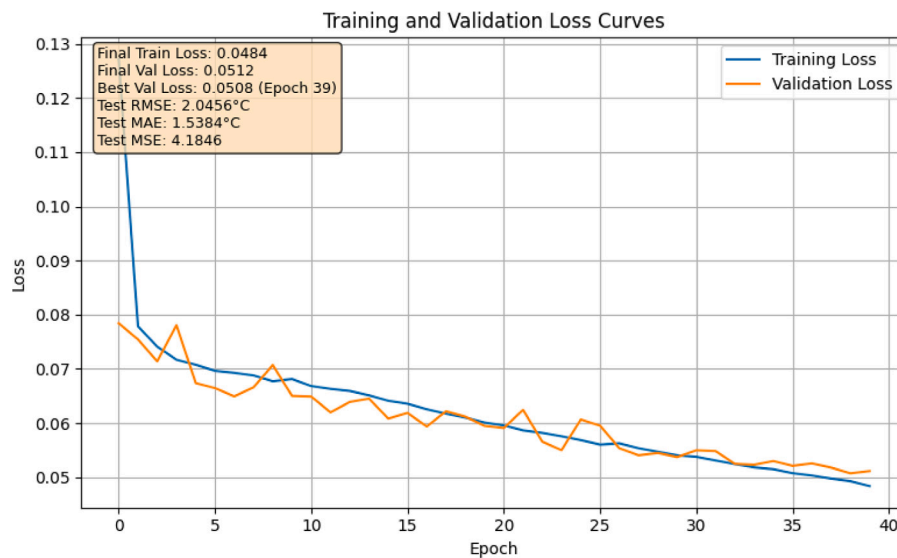


Fig. 13. Training and validation loss curves of the extended U-Net downscaling model (Model 2: $r2m + d2m + \text{elevation}$). Final test metrics (RMSE = 2.05 °C, MAE = 1.54 °C) are reported in the legend, demonstrating improved generalization over the baseline (Fig. 7) and the absence of overfitting.

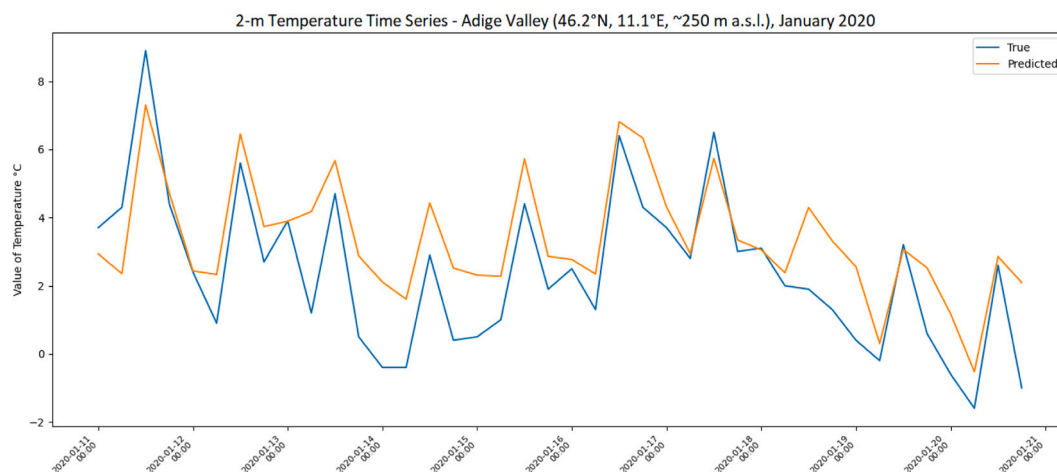


Fig. 14. Time series of observed (ALPINE, blue) and predicted (U-Net, orange) 2-m air temperature at the same representative location as Fig. 8 over a 10-day period (January 2020), illustrating temporal consistency and predictive accuracy of the extended model. The representative location corresponds to pixel ($i = 0, j = 182$) in the ALPINE-TST-250 grid, at approximately 46.2°N, 11.1°E at an elevation of ≈ 250 m a.s.l. (Adige Valley floor, Trentino). Comparing this panel with Fig. 8, the extended model (Model 2) reduces the systematic cold-period underestimation visible in the baseline, consistent with the improved valley-floor temperature representation brought by the $d2m$ predictor.

of meteorological variables and incorporating temporal deep learning techniques to achieve continuous hourly resolution.

CRediT authorship contribution statement

Ihcene Djouama: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Nabil Kadache:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Rachid Seghir:** Writing – review & editing, Validation, Supervision.

Ethics approval

Ethics approval was not required for this study, as it did not involve human participants, human data, or animals.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors confirm that they used generative artificial intelligence (AI) tools, specifically ‘Gemini’, to assist with **language editing and grammar checking** of the final manuscript draft. The authors wrote all scientific content, conceptualized the research, performed the analysis, and are fully responsible for the integrity and precision of the content presented.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Software and data availability

The ERA5-Land reanalysis data used in this study is publicly available from the Copernicus Climate Change Service (C3S) Climate Data Store (CDS). Available at: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=download>. The ALPINE-TST-250 high-resolution dataset was downloaded by the authors on 30/05/2025. Available at: <https://zenodo.org/records/14795887>.

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